PEARSON EDEXCEL INTERNATIONAL GCSE

Mathematics A

Unit 2 • Chapter OL-T37 Classified

Cross-session • chapter-organised candidate

25 classified items • 45 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

Contents

Unit 2 • Unit 2 • Chapter OL-T37 • 25 items • 45 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Transformations of Graphs

Unit 2 • 45 marks • 25 item(s)

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

19 The equation of a curve is $y = f(x)$

There is only one minimum point on the curve.
The coordinates of this minimum point are (8, –12)

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x) + 3$

(.......................... , ...)
(1)

(ii) $y = f(2x)$

(.......................... , ...)
(1)

(Total for Question 19 is 2 marks)

WORKING SPACE

Question 09
4MA1-1H Q21

MODULAR UNIT 2
2 marks

21 A curve has equation $y = \mathrm{f}(x)$

There is one minimum point on this curve.
The coordinates of this minimum point are (5, –4)

Write down the coordinates of the minimum point on the curve with equation

(i) $y = \mathrm{f}(x + 7)$

(.......................... , ...)
(1)

(ii) $y = \mathrm{f}(x) - 6$

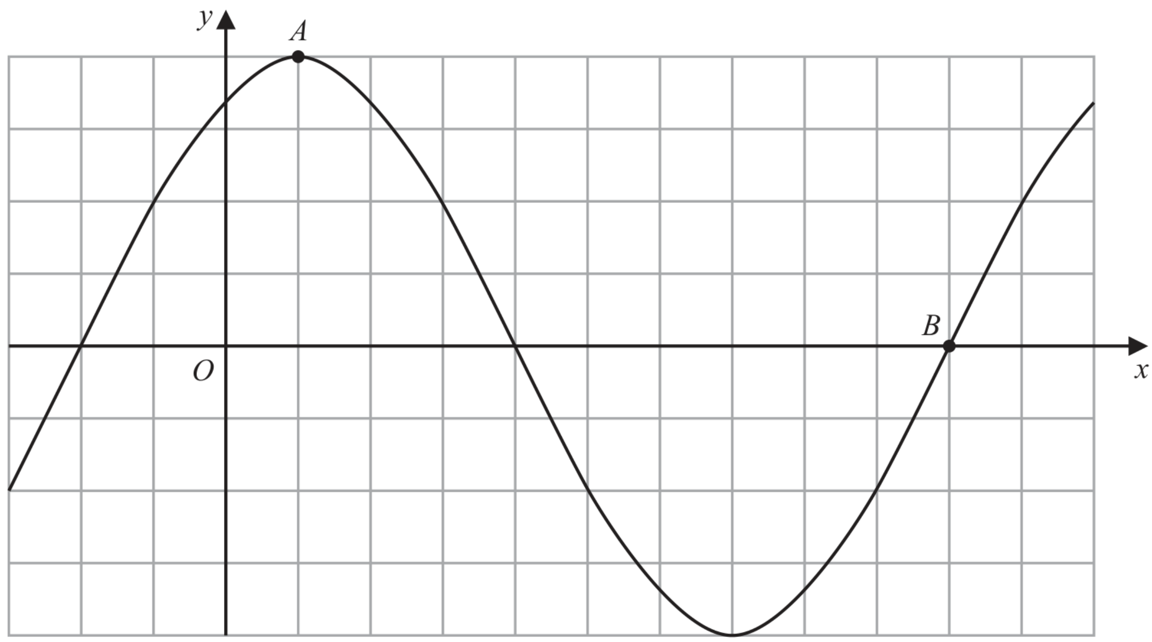
(.......................... , ...)
(1)

WORKING SPACE

Question 27
4MA1-2H Q25

MODULAR UNIT 2
2 marks

25 The diagram shows a sketch of the graph of $y = 2\sin(x + 60)°$



(i) Find the coordinates of the point A

(.......................... , ...)
(1)

(ii) Find the coordinates of the point B

(.......................... , ...)
(1)

Working space for Question 27

Continue your method

UNIT 2
2 marks

WORKING SPACE

Question 19

4MA1-1H Q24 (a)

MODULAR UNIT 2
2 marks

24 The curve with equation $f(x) = 5x^2 + 9x + 2$ is transformed to the curve with equation

$g(x) = 5(x+4)^2 + 9(x+4) + 8$ by the translation $\begin{pmatrix} a \\ b \end{pmatrix}$

(a) Write down the value of a and the value of b

$a =$ ....

$b =$ ....

(2)

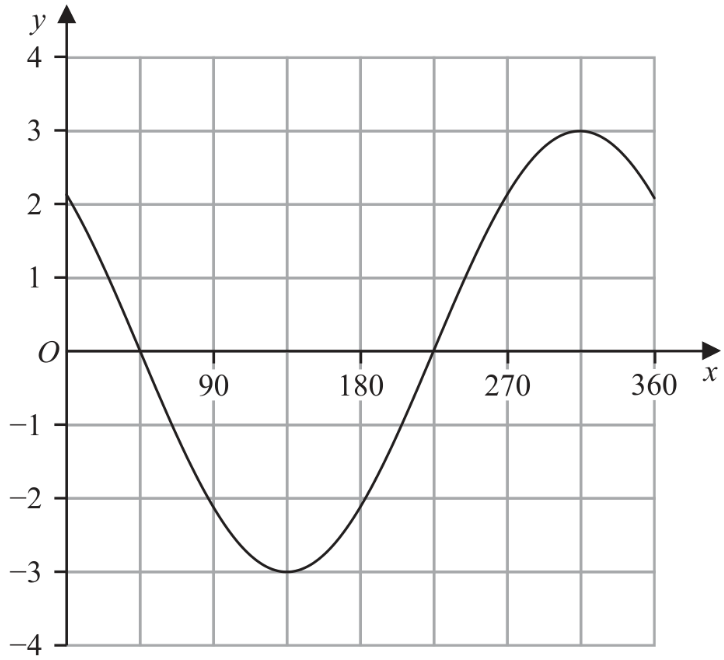
WORKING SPACE

Question 20

4MA1-1H Q24 (b)

MODULAR UNIT 2
2 marks

The graph of $y = p\cos(x+q)^\circ$ for $0 \leqslant x \leqslant 360$ is drawn on the grid below.



Given that $p > 0$ and $0 < q < 360$

(b) find the value of p and the value of q

p = ...

q = ...

(2)

WORKING SPACE

Working space for Question 20

Continue your method

UNIT 2
2 marks

WORKING SPACE

Question 17
4MA1-1H Q21

MODULAR UNIT 2
2 marks

21 A curve has equation $y = f(x)$

There is only one turning point on the curve.
The coordinates of this turning point are (6, 5)

Write down the coordinates of the turning point on the curve with equation

(a) $y = f(x - 4)$

(.............................. , ...)
(1)

(b) $y = f(3x)$

(.............................. , ...)
(1)

WORKING SPACE

Question 30

4MA1-2H Q19 M01

MODULAR UNIT 2
1 marks

19 A curve has equation $y = f(x)$

There is only one minimum point on the curve.
The coordinates of this minimum point are (5, 4)

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x + 5)$

(.............................. , ...)
(1)

WORKING SPACE

Question 31

4MA1-2H Q19 M02

MODULAR UNIT 2

1 marks

19 A curve has equation $y = f(x)$

There is only one minimum point on the curve.
The coordinates of this minimum point are (5, 4)

(ii) $y = 3f(x)$

(.............................. , ...)

(1)

WORKING SPACE

Question 32

4MA1-2H Q19 M03

MODULAR UNIT 2

1 marks

19 A curve has equation $y = f(x)$

There is only one minimum point on the curve.
The coordinates of this minimum point are (5, 4)

(iii) $y = f(x) - 7$

(.............................. , ...)

(1)

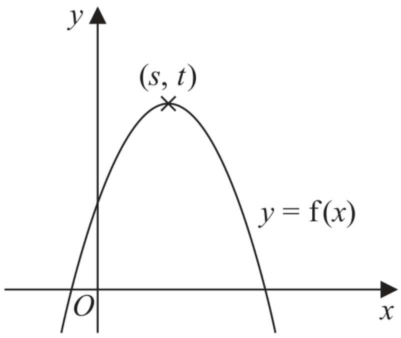
WORKING SPACE

Question 16

4MA1-1H Q20 Q20(i)

MODULAR UNIT 2
1 marks

20 The diagram shows a sketch of part of the curve with equation $y = \mathrm{f}(x)$



There is one maximum point on this curve.
The coordinates of this maximum point are (s, t)

Find, in terms of s and t, the coordinates of the maximum point on the curve with equation

(i) $y = \mathrm{f}(x - 2)$

(.............................. , ...)
(1)

Working space for Question 16

Continue your method

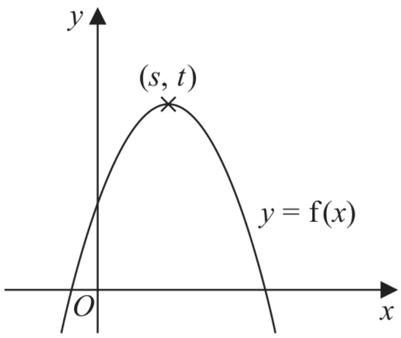
UNIT 2
1 marks

WORKING SPACE

Question 17
4MA1-1H Q20 Q20(ii)

MODULAR UNIT 2
1 marks

20 The diagram shows a sketch of part of the curve with equation $y = \mathrm{f}(x)$



There is one maximum point on this curve.
The coordinates of this maximum point are (s, t)

Find, in terms of s and t, the coordinates of the maximum point on the curve with equation

(ii) $y = 3\mathrm{f}(x)$

(.......................... , ...)
(1)

Working space for Question 17

Continue your method

UNIT 2

1 marks

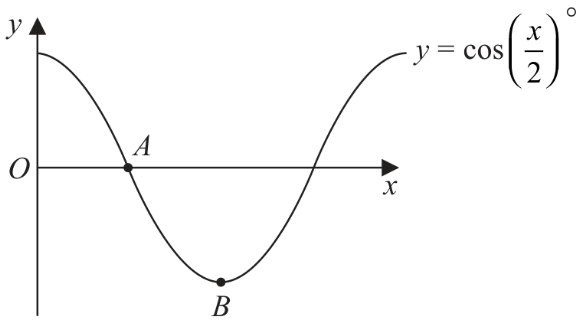
WORKING SPACE

Question 31

4MA1-2H Q23 Q23(i)

MODULAR UNIT 2
1 marks

23 The diagram shows a sketch of the graph of $y = \cos\left(\dfrac{x}{2}\right)^\circ$



(i) Find the coordinates of the point A

(.............................. , ...)

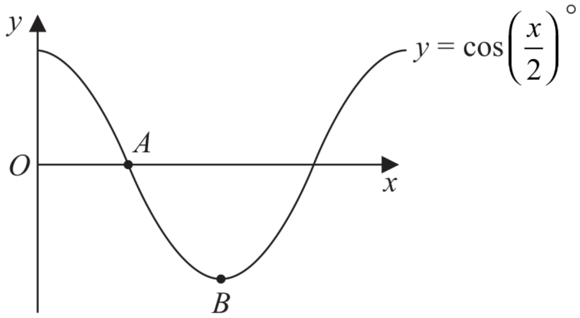
(1)

WORKING SPACE

Question 32
4MA1-2H Q23 Q23(ii)

MODULAR UNIT 2
1 marks

23 The diagram shows a sketch of the graph of $y = \cos\left(\frac{x}{2}\right)^{\circ}$



(ii) Find the coordinates of the point B

(.............................. , ...)
(1)

Working space for Question 32

Continue your method

UNIT 2
1 marks

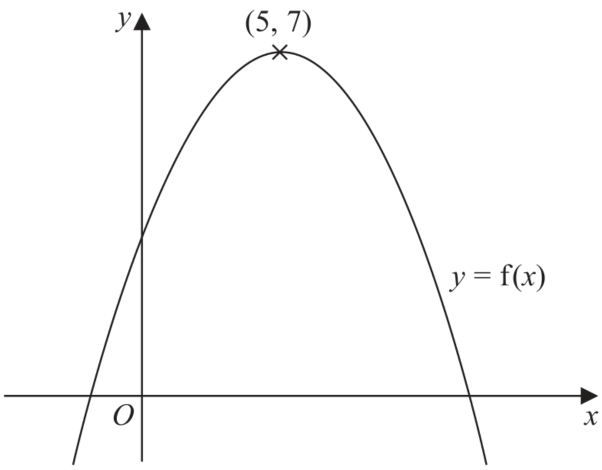
WORKING SPACE

Question 14

4MA1-1H Q23

MODULAR UNIT 2
2 marks

23 The diagram shows a sketch of the curve with equation $y = f(x)$



There is only one maximum point on the curve.
The coordinates of this maximum point are (5, 7)

Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x + 9)$

(...................... , ...)

(ii) $y = f(x) + 3$

(...................... , ...)

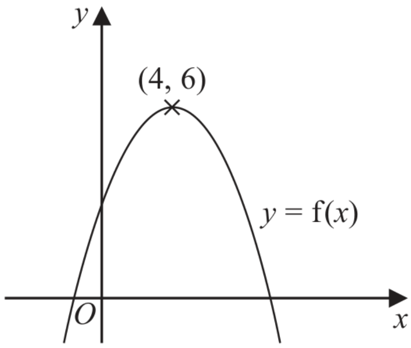
Working space for Question 14

Continue your method

UNIT 2
2 marks

WORKING SPACE

21 The diagram shows a sketch of part of the curve with equation $y = f(x)$



There is one maximum point on this curve.
The coordinates of this maximum point are (4, 6)

(a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x + 4)$

(.............................. , ...)

(ii) $y = f(2x)$

(.............................. , ...)

(2)

YOUR WORKING

The equation of a curve **C** is $y = x^2 + 3x + 4$

The curve **C** is transformed to curve **S** under the translation $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$

(b) Find an equation of curve **S**.

You do not need to simplify the equation.

....

(2)

YOUR WORKING

21 A curve has equation $y = f(x)$

The coordinates of the minimum point on this curve are (–9, 15)

(a) Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x + 3)$

(.......................... , ...)

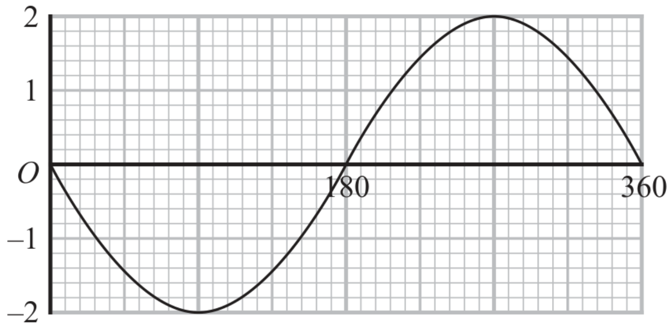
(ii) $y = \frac{1}{3}f(x)$

(.......................... , ...)

(2)

YOUR WORKING

The graph of $y = a\cos(x + b)°$ for $0 \leqslant x \leqslant 360$ is drawn on the grid below.



Given that $a > 0$ and that $0 < b < 360$

(b) find the value of a and the value of b.

$a =$ ....

$b =$ ....

(2)

YOUR WORKING

22 The curve **S** has equation $y = \mathrm{f}(x)$ where $\mathrm{f}(x) = x^2$
The curve **T** has equation $y = \mathrm{g}(x)$ where $\mathrm{g}(x) = 2x^2 - 12x + 13$

By writing $\mathrm{g}(x)$ in the form $a(x - b)^2 - c$, where a, b and c are constants, describe fully a series of transformations that map the curve **S** onto the curve **T**.

....

....

....

YOUR WORKING

21 The curve **C** has equation $y = \mathrm{f}(x)$ where $\mathrm{f}(x) = 9 - 3(x + 2)^2$
The point A is the maximum point on **C**.

(a) Write down the coordinates of A.

(.......................... , ...)

(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

....

(1)

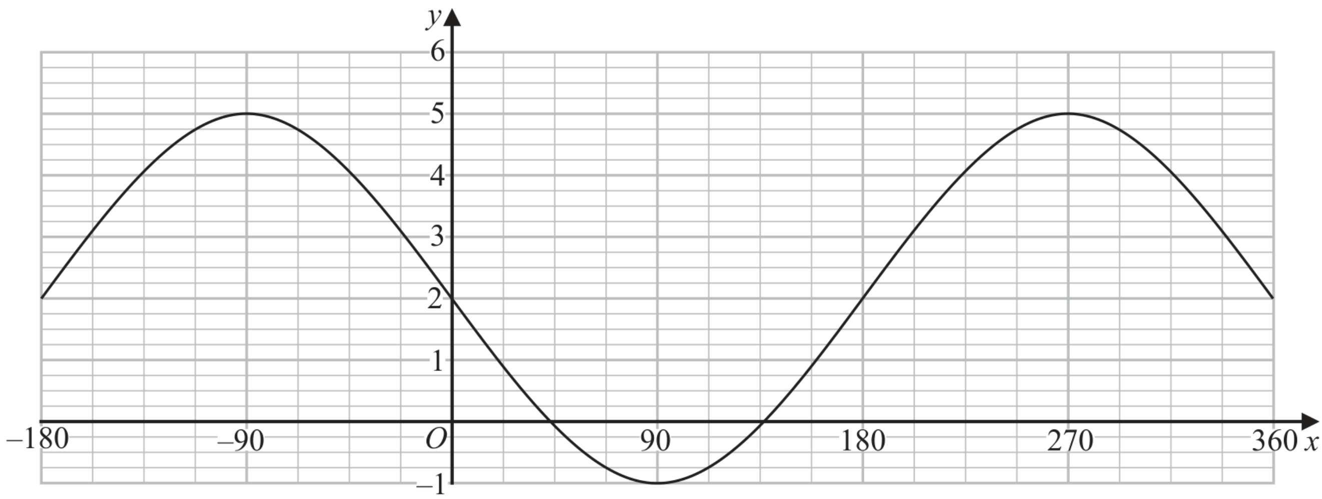
The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

....

(1)

The graph of $y = a\cos(x - b)° + c$ for $-180 \leqslant x \leqslant 360$ is drawn on the grid below.



(d) Find the value of a, the value of b and the value of c.

a = ...

b = ...

c = ...

(3)

YOUR WORKING

21 The curve **C** has equation $y = f(x)$ where $f(x) = 9 - 3(x + 2)^2$
The point A is the maximum point on **C**.

(a) Write down the coordinates of A.

(.............................. , ...)

(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

....

(1)

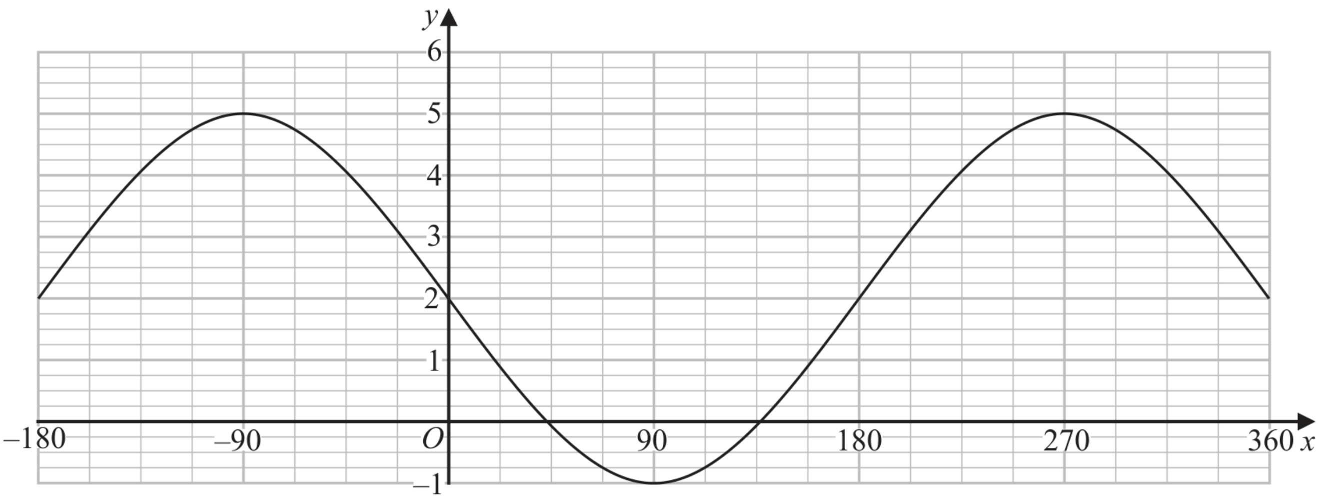
The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

....

(1)

The graph of $y = a\cos(x - b)° + c$ for $-180 \leqslant x \leqslant 360$ is drawn on the grid below.



(d) Find the value of a, the value of b and the value of c.

a = ...

b = ...

c = ...

(3)

YOUR WORKING

November 2021 | 4MA1-2H | Question 21 | 1 marks

Chapter OL-T37: Transformations of Graphs

21 The curve **C** has equation $y = \mathrm{f}(x)$ where $\mathrm{f}(x) = 9 - 3(x + 2)^2$

The point A is the maximum point on **C**.

(a) Write down the coordinates of A.

(.......................... , ...)

(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

....

(1)

The curve **C** is transformed to the curve **T**.

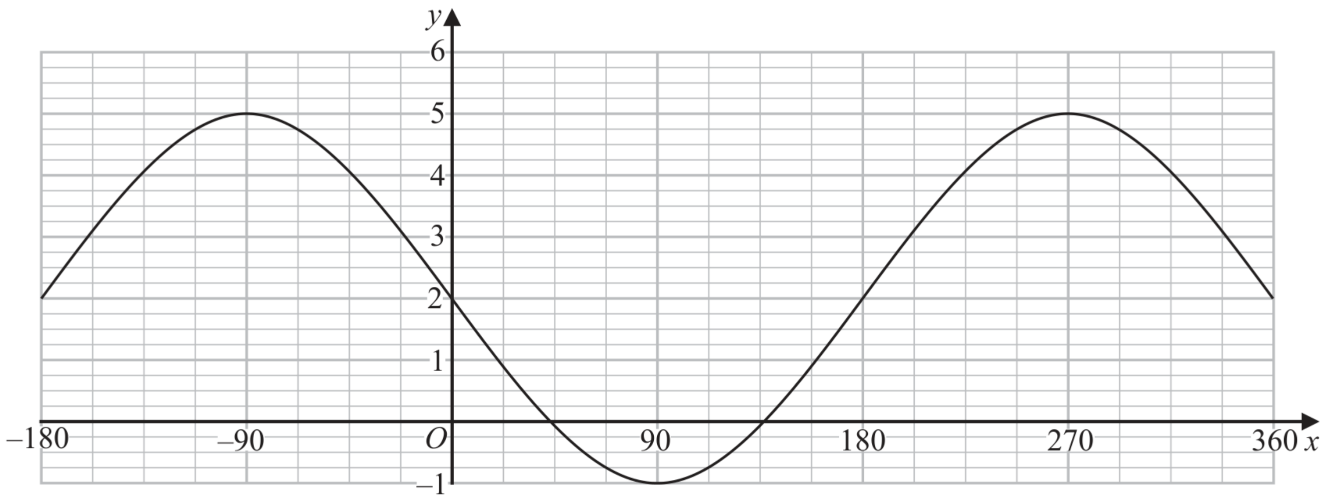
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

....

(1)

The graph of $y = a\cos(x - b)° + c$ for $-180 \leqslant x \leqslant 360$ is drawn on the grid below.



(d) Find the value of a, the value of b and the value of c.

a = ...

b = ...

c = ...

(3)

YOUR WORKING

21 The curve **C** has equation $y = \mathrm{f}(x)$ where $\mathrm{f}(x) = 9 - 3(x + 2)^2$
The point A is the maximum point on **C**.

(a) Write down the coordinates of A.

(.......................... , ...)
(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

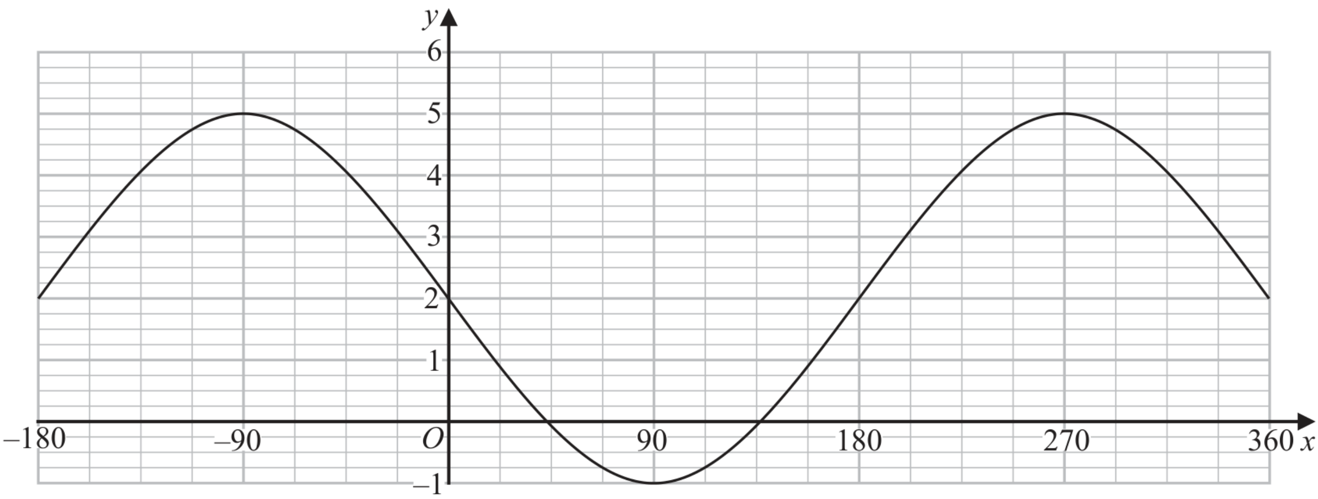
....
(1)

The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

....
(1)

The graph of $y = a\cos(x - b)° + c$ for $-180 \leqslant x \leqslant 360$ is drawn on the grid below.



(d) Find the value of a, the value of b and the value of c.

a = ...

b = ...

c = ...
(3)

YOUR WORKING

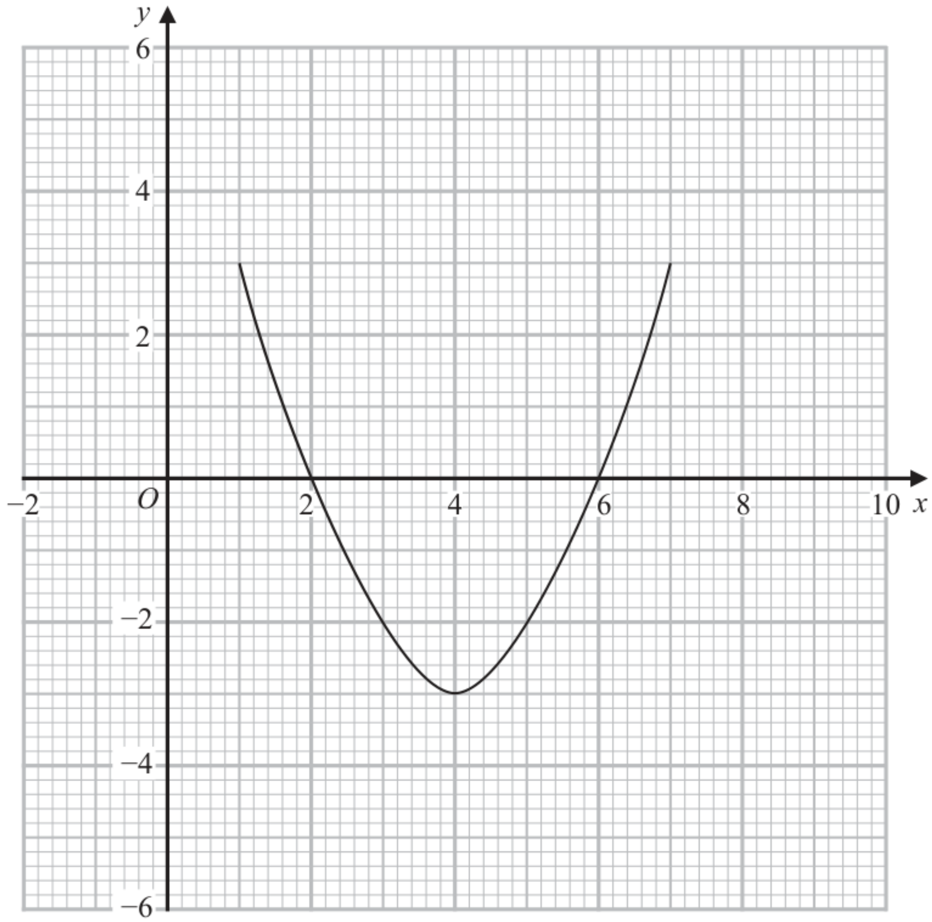
25 The curve with equation $y = \mathrm{g}(x)$ is transformed to the curve with equation $y = -\mathrm{g}(x)$ by the single transformation **T**.

(a) Describe fully the transformation **T**.

....

(1)

The diagram shows the graph of $y = \mathrm{f}(x)$



(b) On the grid, draw the graph of $y = 2\mathrm{f}(x - 1)$

(2)

(Total for Question 25 is 3 marks)

YOUR WORKING

24 A curve has equation $y = f(x)$

There is only one turning point on the curve.
The coordinates of this turning point are (4, 3)

Write down the coordinates of the turning point on the curve with equation

(i) $y = f(x + 5)$

(.......................... , ...)
(1)

(ii) $y = f(x) + 7$

(.......................... , ...)
(1)

(iii) $y = f(2x)$

(.......................... , ...)
(1)

(Total for Question 24 is 3 marks)

YOUR WORKING